Image Enhancement for High-Resolution Visual Contents

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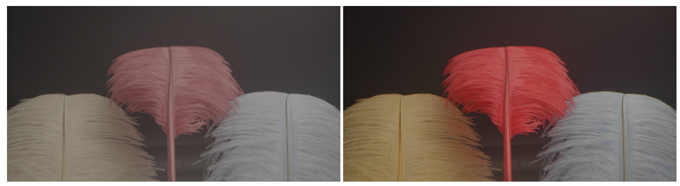
***Abstract*—This paper proposes an image enhancement method using gamma neural networks and exponential transformation. When acquiring an image, degradation occurs in very many imaging systems, and the quality of the image acquired by surrounding environmental factors decreases due to the combination of deteriorating elements. Alternatively, work that facilitates post-treatment may be performed by artificially deteriorating for post-treatment directly. However, if the information on these additional tasks is not known, there is a problem that the post-processing process is expensive or additional degradation occurs. To solve this problem, this paper uses a neural network that estimates gamma maps through residual learning for images that require post-processing, and finally applies exponential transformations to perform contrast improvement. The contrast improvement method proposed through the experimental results provides an image with less color distortion compared to the existing method.**

***Index Terms*—image enhancement, neural network, visual contents**

I. Introduction

Recently, high-quality images have been continuously used in fields such as vehicle image storage devices, autonomous driving, and CCTV. At this time, since the image acquired in a low-contrast environment degraded in contrast and information characteristics, an image enhancement method for obtaining high-quality images is required [1]–[3]. Histogram-based methods have been commonly used as a contrast improvement method. Histogram-based contrast improvement improves the image by obtaining a cumulative distribution function for image pixel values and then relocating pixel values to an even distribution. In addition, retinex theory based on human visual system analysis improves contrast by eliminating lighting components.

In recent, a method using convolutional neural networks is widely used, and a method of separating and recombining lighting and reflective components using retinex theory is used. However, there is a problem that the contrast improvement methods through various approaches fundamentally cause additional degradation due to color distortion and signal saturation. To solve this problem, this paper proposes an image improvement method using gamma neural networks. The proposed method combines gamma maps and input images estimated via residual learning, convolutional neural networks, and then performs exponential transformation to finally obtain improved images. As shown in Fig.1, As a result of the experiment, it was possible to obtain a high-quality contrast enhancement image with less degradation compared to the existing method.



(a) (b)

Fig. 1. Proposed method Result (a) Input image, (b) Proposed method result

II. Proposed method

Several degradations occur when high-resolution images are acquired by the imaging system, or there is a process of artificially adding log-transformation processes for post-processing. Since log transformation uses a different log for each imaging system, metadata is required in the post-processing process. However, there is a problem that additional costs occur in the post-processing process when metadata is unknown. To solve this problem, the proposed method proposes an image improvement method using a convolution neural network that estimates gamma maps that estimate arbitrary log-transformation values.

A. Gamma Convolutional Neural Networks

Since the proposed gamma convolutional neural network does not know the unknown log-transformation coefficient, the proposed method calculates the log image using explicit transformation, and adds the estimated gamma map to finally obtain the resulting image. This process is performed because the log coefficient used in the log-transformation process is unknown, and, as shown in Fig. 1, an improved image can be obtained with a simple explicit transformation, but further degradation occurs due to unknown log coefficients. For this reason, we estimate the gamma map to correct the outcome of the exposure transformation through the gamma convolutional neural network. The proposed gamma convolutional neural network consists of 10 convolution and

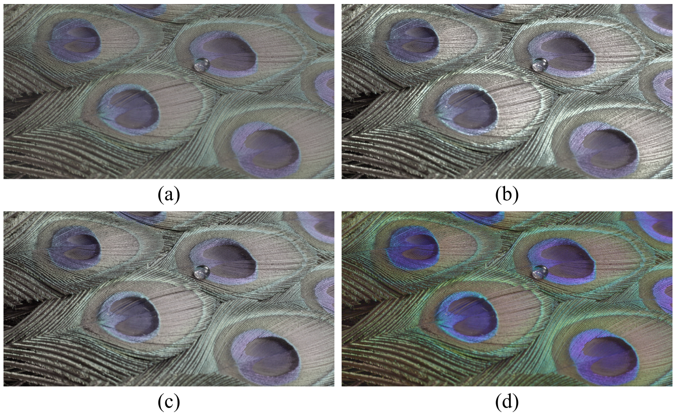


Fig. 2. Comparitive results of image enhancement methods. (a) Input image (NIQE: 3.0188), (b) GLCE [6] (NIQE: 3.1987), (c) CVC [5] (NIQE: 2.9929), (d) Proposed method (NIQE: 2.9387).

ReLU block structures for processing high-resolution images, and the network equation is followed as [4]

$$\hat{y} = f(x) + x, \tag{1}$$

where $\hat{y}$ is gamma map, $f(\cdot)$ the gamma convolutional neural networks, and x is an input image. The $\hat{y}$ gamma map, estimated through the gamma convolutional neural network, calculates arbitrary log transformation through the input image x and adds the gamma map to obtain an enhanced image.

B. Loss function

In the proposed method, the loss function is calculated using mean absolute error (MAE), and the equation is followed as

$$\ell_1 = \|y - (\exp(x) + f(x) + x)\|_1^1, \tag{2}$$

where ℓ_1 is the MAE loss function for gamma convolutional neural network learning and y is the ground truth image.

III. Experimental Results

The proposed method compared performance using GLCE, and CVC methods, and natural image quality evaluator (NIQE). The lower the NIQE, the more similar the result to the natural images. As shown in Fig. 2, existing improvement methods such as GLCE, CVC, and methods show improved results, but there is a problem that information loss occurs due to signal saturation. On the other hand, the proposed method shows improved results and high PSNR and SSIM figures without color distortion compared to the conventional method, and shows that high-resolution visual content image improvement is possible.

IV. Conculusion

This paper proposes a method for image enhancement through gamma neural networks. The proposed method estimates improvements using exponential transformations and gamma maps via gamma neural networks for images with added artificial degradation or images requiring improvement. Using the estimated gamma map, we add to the exponentially transformed results to reduce signal saturation and finally obtain quality improvement results. We show that high-quality image enhancement results can be provided by applying the proposed method to high-resolution visual content through experimental results.

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References

[1] S. Ahn, J. Shin, H. Lim, J. Lee, and J. Paik, "CODEN: combined optimization-based decomposition and learning-based enhancement network for Retinex-based brightness and contrast enhancement," Opt. Express 30, 23608-23621 (2022)

[2] J. Shin, H. Park and J. Paik, "Region-Based Dehazing via Dual-Supervised Triple-Convolutional Network," in IEEE Transactions on Multimedia, vol. 24, pp. 245-260, 2022.

[3] H. Lim, S. Yu, K. Park, D. Seo and J. Paik, "Texture-Aware Deblurring for Remote Sensing Images Using l0-Based Deblurring and l2-Based Fusion," in IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, vol. 13, pp. 3094-3108, 2020.

[4] Kim, J., Lee, J. K., Lee, K. M. (2016). Accurate image super-resolution using very deep convolutional networks. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 1646-1654).

[5] T. Celik and T. Tjahjadi, "Contextual and Variational Contrast Enhancement," in IEEE Transactions on Image Processing, vol. 20, no. 12, pp. 3431-3441, Dec. 2011.

[6] Y. Zhu, X. Fu and A. Liu, "Learning Dual Transformation Networks for Image Contrast Enhancement," in IEEE Signal Processing Letters, vol. 27, pp. 1999-2003, 2020.